



2002
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved scientific calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt all questions 1-7
- All questions are of equal value
- Start a new writing booklet for each question

Total marks (84)

Attempt Questions 1-7

All questions are of equal value.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)	Use a SEPARATE Writing Booklet.	Marks
(a) Evaluate $\cos\left[\tan^{-1}\left(-\sqrt{3}\right)\right]$. Show all working.		2
(b) Solve the inequality $\frac{x}{2x-1} > 0$.		3
(c) If α, β, γ are the roots of the equation $2x^3 - 6x^2 + x + 2 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.		3
(d) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{6}} \frac{dx}{4+2x^2}$.		4

Question 2 (12 marks)

Use a SEPARATE Writing Booklet.

Marks

- (a) Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \tan^2 x \, dx$ using the substitution $u = \tan x$. **3**
- (b) Taking $\frac{\pi}{2}$ as your first approximation, apply Newton's method once to find a better approximation to the root of the equation $3 \sin x - 2x = 0$. **3**
- (c) Differentiate $y = \tan^{-1}(\log_e x)$. **2**
- (d) A committee of six people is to be chosen from 8 men and 6 women. **4**
What is the probability that at least 3 men and 2 women will be chosen for this committee?

Question 3 (12 marks)

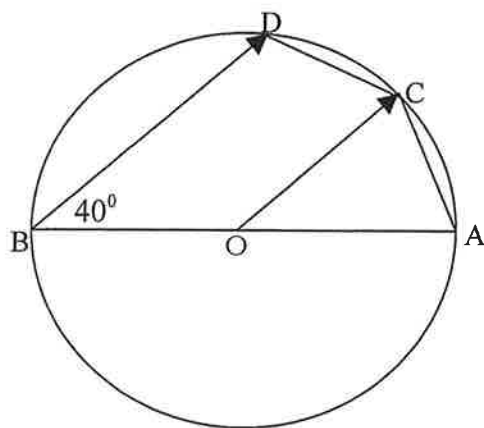
Use a SEPARATE Writing Booklet.

Marks

(a) $P(x) = x^3 + x^2$

- i) Sketch the polynomial $P(x)$ indicating all x and y intercepts. 2
- ii) The tangent at $A(1, 2)$ to the curve $y = x^3 + x^2$ meets the curve again at B .
Draw this on your diagram in part (i).
Use calculus to show that AB has equation $y = 5x - 3$. 2
- iii) Show that the tangent AB and the curve $y = x^3 + x^2$ meet where $x^3 + x^2 - 5x + 3 = 0$. 1
- iv) Hence or otherwise, find the coordinates of B . 3

(b)



4

$ABDC$ is a cyclic quadrilateral. AB is a diameter of the circle with centre O . BD and OC are parallel, and $\angle OBD = 40^\circ$. C and D are joined.

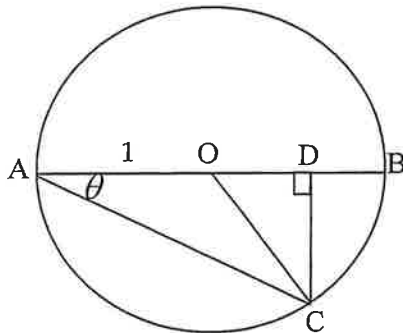
- i) Copy this diagram into your writing booklet.
- ii) Prove that OC bisects $\angle ACD$, giving geometric reasons for your answer.

Question 4 (12 marks)

Use a SEPARATE Writing Booklet.

Marks

a)



AB is a diameter of a circle centre O , radius 1 unit. CD is perpendicular to AB . $\angle OAC = \theta$.

- i) Explain why $\angle DOC = 2\theta$ giving geometrical reasons. 2
 - ii) Show that the perimeter P of $\triangle DOC$ is given by $P = 1 + \sin 2\theta + \cos 2\theta$. 2
 - iii) Find the value of R and α , such that $\sin 2\theta + \cos 2\theta = R \sin(2\theta + \alpha)$. 3
 - iv) Hence or otherwise, find the maximum value of the perimeter P of $\triangle DOC$, and the corresponding value of θ . 2
- (b) A machine manufacturing clone soldiers for the Republic is known to produce on average, one defective clone soldier per hundred. 3
- What is the probability that just one defective clone soldier will be made in the manufacture of 10 such clone soldiers ?
- Give your answer correct to 3 decimal places.

Question 5 (12 marks) Use a SEPARATE Writing Booklet.

Marks

(a) Consider the function $f(x) = x^2 - 2x$.

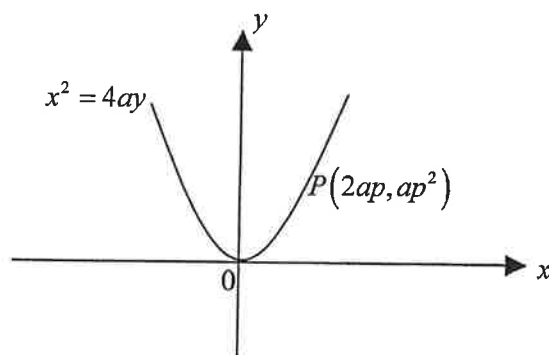
i) Find the domain over which $f(x) = x^2 - 2x$ is monotonic increasing.

1

ii) Find the inverse function $f^{-1}(x)$ over this restricted domain.

3

(b) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$.



i) Show that the equation of the normal to the curve of the parabola at the point P is $x + py = 2ap + ap^3$.

2

ii) Find the coordinates of the point Q where the normal at P meets the y -axis.

1

iii) Show that the coordinates of the point R , which divides PQ externally in the ratio 2:1 are given by $(-2ap, 4a + ap^2)$.

2

iv) Find the cartesian equation of the locus of R and describe the locus in geometrical terms.

3

Question 6 (12 marks)	Use a SEPARATE Writing Booklet.	Marks
(a) Let $f(x) = \frac{x}{x^2 - 1}$.		
i) For what values of x is $f(x)$ undefined ?		1
ii) Show that $y = f(x)$ is an odd function.		2
iii) Show that $f'(x) < 0$ at all values of x for which the function is defined.		2
iv) Hence sketch $f(x)$ showing all important features.		2
(b) Find the term independent of x in the expansion $\left(x^3 + \frac{2}{x}\right)^8$.		3
(c) A boat seats 8 people and has 4 seats on the stroke side and 4 seats on the bow side. In how many ways can 8 oarsmen be seated if 3 can row only on the stroke side and 2 can only row on the bow side ?		2

Question 7 (12 marks) Use a SEPARATE Writing Booklet. **Marks**

(a) By considering the expansion $(1+x)^n$, show that : **4**

$${}^nC_1 + 2.{}^nC_2 + 3.{}^nC_3 + 4.{}^nC_4 + \dots + n.{}^nC_n = n.2^{n-1}$$

(b) The region bounded by the curve $y = \cos 2x$ and the x-axis between $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x-axis. Find the exact volume of the solid of revolution formed. **4**

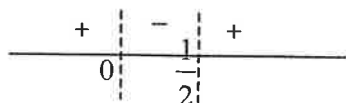
(c) Use mathematical induction to prove that for all positive integers n : **4**

$$\sum_{r=1}^n r(r!) = (n+1)! - 1.$$

Question 1.

$$\begin{aligned} \text{a) } \cos\left[\tan^{-1}(-\sqrt{3})\right] &= \cos\left[-\tan^{-1}\sqrt{3}\right] \\ &= \cos\left(-\frac{\pi}{3}\right) \quad \checkmark \\ &= \cos\frac{\pi}{3} \\ &= \frac{1}{2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) } \frac{x}{2x-1} > 0 \quad x \neq \frac{1}{2} \\ x < 0 \quad \text{or} \quad x > \frac{1}{2} \quad \checkmark\checkmark\checkmark \end{aligned}$$



$$\begin{aligned} \text{c) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \quad \checkmark \\ &= \frac{1}{-1} \quad \checkmark \\ &= -\frac{1}{2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{d) } \int_{\sqrt{2}}^{\sqrt{6}} \frac{dx}{4+2x^2} &= \frac{1}{2} \int_{\sqrt{2}}^{\sqrt{6}} \frac{dx}{2+x^2} \\ &= \frac{1}{2} \times \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{x}{\sqrt{2}} \right]_{\sqrt{2}}^{\sqrt{6}} \quad \checkmark \\ &= \frac{1}{2\sqrt{2}} \left[\tan^{-1}\sqrt{3} - \tan^{-1}1 \right]_{\sqrt{2}}^{\sqrt{6}} \quad \checkmark \\ &= \frac{1}{2\sqrt{2}} \left[\frac{\pi}{3} - \frac{\pi}{4} \right] \quad \checkmark \\ &= \frac{\pi}{24\sqrt{2}} \quad \text{or} \quad \frac{\pi\sqrt{2}}{48} \quad \checkmark \end{aligned}$$

Question 2.

a) $\int_0^{\pi} \sec^2 x \tan^2 x dx$ $u = \tan x, du = \sec^2 x dx$

$x = 0, u = 0$ ☒

$x = \frac{\pi}{4}, u = 1$

$\int_0^1 u^2 du = \left[\frac{u^3}{3} \right]_0^1$ ☒

$= \frac{1}{3}$ ☒

b) $f\left(\frac{\pi}{2}\right) = 3 - \pi$ ☒

$f'\left(\frac{\pi}{2}\right) = 3 \cos \frac{\pi}{2} - 2 = -2$ ☒

$x_1 = \frac{\pi}{2} - \frac{3 - \pi}{-2}$
 $= \frac{3}{2}$ ☒

c) $\frac{dy}{dx} = \frac{1}{1 + (\log_e x)^2} \times \frac{1}{x}$ ☒ ☒

$= \frac{1}{x(1 + 2 \log_e x)} \text{ or } \frac{1}{x(1 + \log_e x^2)}$

d) $P(\text{at least } 3M2W) = P(3M3W) \text{ or } P(4M2W)$

$P(3M3W) = \frac{{}^8C_3 \times {}^6C_3}{{}^{14}C_6}$ ☒ ☒

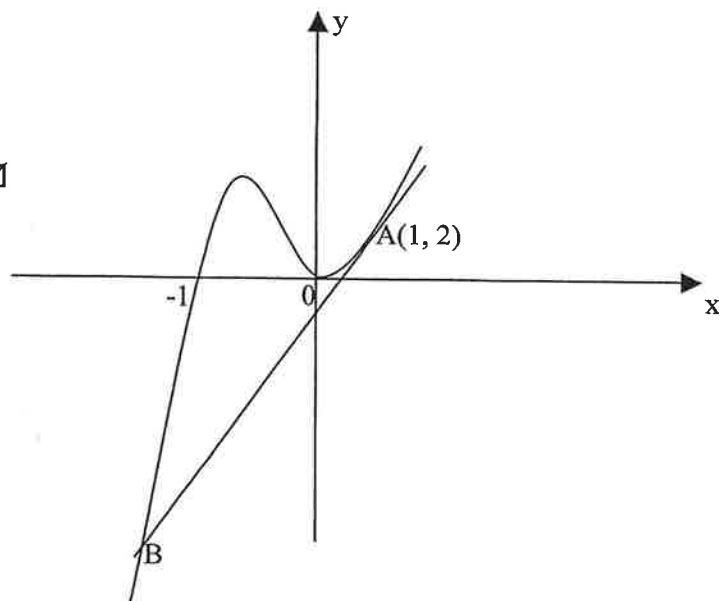
$P(4M2W) = \frac{{}^8C_4 \times {}^6C_2}{{}^{14}C_6}$ ☒

$\therefore P(\text{at least } 3M2W) = \frac{1120}{3003} + \frac{1050}{3003}$
 $= \frac{2170}{3003}$
 $= \frac{310}{429}$ ☒

Question 3

a)(i)

☑☑



(ii) $P(x) = x^2(x+1)$

$P'(x) = 3x^2 + 2x$

$P'(1) = 5$ ☑

$y - 2 = 5(x - 1)$

$y = 5x - 3$ ☑

(iii) $y = 5x - 3$ and $y = x^3 + x^2$

$5x - 3 = x^3 + x^2$ ☑

$\therefore x^3 + x^2 - 5x + 3 = 0$

(iv) $x = 1$ is a solution, $(x - 1)$ is a factor

$$\begin{array}{r} x^2 + 2x - 3 \\ x-1 \overline{) x^3 + x^2 - 5x + 3} \end{array} \quad \text{☑}$$

$$\begin{array}{r} x^3 - x^2 \\ \underline{ 2x^2 - 5x} \\ 2x^2 - 2x \\ \underline{ -3x + 3} \end{array}$$

$x^3 + x^2 - 5x + 3 = (x - 1)^2(x + 3)$ ☑

$\therefore x = -3 \quad y = -18$

$B(-3, -18)$ ☑

b) $\angle COA = 40^\circ$ (corresponding $\angle$'s; $BD \parallel OC$) ☑

$\angle OCA = 70^\circ$ (angle sum of isos. $\triangle OCA$; $OC = OA$ equal radii) ☑

$\angle DCA = 140^\circ$ (opp. $\angle$'s of cyclic quad. $BDCA$) ☑

$\angle DCO = 140^\circ - 70^\circ$ (adjacent $\angle$'s)

$\therefore \angle DCO = \angle OCA = 70^\circ$ ☑

and hence OC bisects $\angle ACD$.

Question 4

- a) (i) $\angle OCA = \theta$ (equal $\angle$'s of isos. $\triangle OAC$; $OA = OC$ equal radii) ☒
 $\angle DOC = 2\theta$ (exterior $\angle$ of $\triangle OAC$ = sum of opposite 2 interior $\angle$'s) ☒

or

angle at circumference is half the angle at the centre subtended by the same arc BC.

- (ii) $OC = 1$ (radius)

$$\frac{DC}{1} = \sin 2\theta \quad DC = \sin 2\theta \quad \checkmark$$

$$\frac{OD}{1} = \cos 2\theta \quad OD = \cos 2\theta$$

$$P = 1 + \sin 2\theta + \cos 2\theta \quad \checkmark$$

- (iii) $R \sin(2\theta + \alpha) = R \sin 2\theta \cos \alpha + R \cos 2\theta \sin \alpha \quad \checkmark$

$$R = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \checkmark$$

$$\tan \alpha = 1, \alpha = \frac{\pi}{4} \quad \checkmark$$

$$\therefore \sin 2\theta + \cos 2\theta = \sqrt{2} \sin\left(2\theta + \frac{\pi}{4}\right)$$

- (iv) $\sin 2\theta + \cos 2\theta = \sqrt{2} \sin\left(2\theta + \frac{\pi}{4}\right)$ has a max value of $\sqrt{2}$ when:

$$\sin\left(2\theta + \frac{\pi}{4}\right) = 1$$

$$2\theta + \frac{\pi}{4} = \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{8}$$

☒

and maximum value of P is $1 + \sqrt{2} \quad \checkmark$

- b) $P(\text{defective}) = p = 0.01$

$$P(\text{not defective}) = q = 0.99$$

$$P(1 \text{ def. } 9 \text{ not def.}) = {}^{10}C_1 p q^9 \quad \checkmark$$

$$= 10 \times 0.01 \times 0.99^9 \quad \checkmark$$

$$= 0.091 (3d.p.) \quad \checkmark$$

Question 5

(a)

i) $x > 1$ ✓

ii) $x = y^2 - 2y$

$x + 1 = (y - 1)^2$ ✓

$y = 1 \pm \sqrt{x + 1}$ ✓

choose $y > 1$; $\therefore y = 1 + \sqrt{x + 1}$ ✓

(b)

i) $\frac{dy}{dx} = \frac{x}{2a}$; at $x = 2ap$, $\frac{dy}{dx} = p$

$m_N = \frac{-1}{p}$ at $(2ap, ap^2)$ ✓

$y - ap^2 = \frac{-1}{p}(x - 2ap)$

$x + py = 2ap + ap^3$ ✓

ii) $x = 0$ $py = 2ap + ap^3$

$y = 2a + ap^2$

$Q(0, 2a + ap^2)$ ✓

iii) $k : l = 2 : -1$

$\left(\frac{2(0) - 2ap}{2 - 1}, \frac{2(2a + ap^2) - ap^2}{2 - 1} \right)$ ✓✓

$(-2ap, 4a + ap^2)$

iv) $x = -2ap$; $p = \frac{-x}{2a}$

$y = 4a + ap^2$

$y = 4a + a\left(\frac{-x}{2a}\right)^2$ ✓

$y = 4a + \frac{x^2}{4a}$

$4a(y - 4a) = x^2$ ✓

parabola vertex $(0, 4a)$, focal length a i.e. focus $(0, 5a)$ ✓

Question 6

(a)

i) $x = \pm 1$ ☒

ii) $f(-x) = \frac{-x}{(-x)^2 - 1}$

$= \frac{-x}{x^2 - 1}$ ☒

$= -f(x)$

$f(-x) = -f(x)$ hence $f(x)$ is odd. ☒

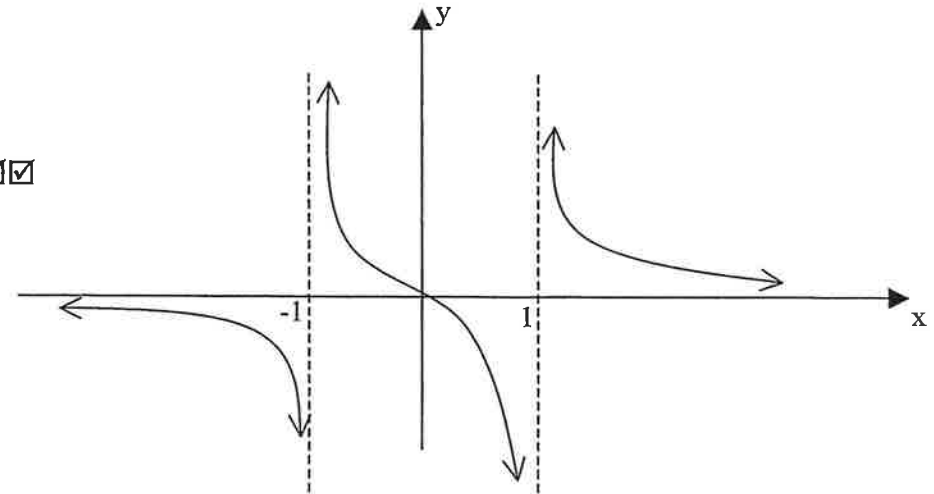
iii) $f'(x) = \frac{(x^2 - 1) \cdot 1 - x(2x)}{(x^2 - 1)^2}$

$= -\frac{(x^2 + 1)}{(x^2 - 1)^2}$ ☒

Now $x^2 + 1 > 0$ and $(x^2 - 1)^2 > 0; x \neq \pm 1$ ☒

Hence $-\frac{x^2 + 1}{(x^2 - 1)^2} < 0$

☒ ☒



(b)

$$\begin{aligned} T_{k+1} &= {}^8C_k (x^3)^{8-k} \left(\frac{2}{x}\right)^k \\ &= {}^8C_k x^{24-3k} (2)^k x^{-k} \\ &= {}^8C_k 2^k x^{24-4k} \end{aligned}$$

$24 - 4k = 0$

$k = 6$ ☒

$$\begin{aligned} T_7 &= {}^8C_6 2^6 \\ &= 1792 \end{aligned}$$

(c)

${}^4P_3 \times {}^4P_2 \times {}^3P_3$ ☒

$= 1728 \text{ ways}$ ☒

Question 7

(a)

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_r x^r + \dots + {}^nC_n x^n \quad \checkmark$$

Differentiate both sides with respect to x

$$n(1+x)^{n-1} = {}^nC_1 + 2{}^nC_2x + 3{}^nC_3x^2 + \dots + r{}^nC_r x^{r-1} + \dots + n{}^nC_n x^{n-1} \quad \checkmark \checkmark$$

let $x = 1$

$$n(2)^{n-1} = {}^nC_1 + 2{}^nC_2 + 3{}^nC_3 + \dots + r{}^nC_r + \dots + n{}^nC_n \quad \checkmark$$

(b)

$$V = \pi \int_0^{\frac{\pi}{4}} (\cos 2x)^2 dx \quad \checkmark$$

$$\cos^2 2x = \frac{1}{2}(\cos 4x + 1)$$

$$V = \frac{\pi}{2} \int_0^{\frac{\pi}{4}} \cos 4x + 1 dx \quad \checkmark$$

$$= \frac{\pi}{2} \left[\frac{\sin 4x}{4} + x \right]_0^{\frac{\pi}{4}} \quad \checkmark$$

$$= \frac{\pi}{2} \left[\left(\frac{\sin \pi}{4} + \frac{\pi}{4} \right) - \left(\frac{\sin 0}{4} + 0 \right) \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} \right]$$

$$= \frac{\pi^2}{8} \quad \checkmark$$

(c)

Test for $n = 1$

$$\text{LHS} = 1(1)! \quad \text{RHS} = 2! - 1$$

$$= 1 \quad = 1$$

$$\text{LHS} = \text{RHS and it is true for } n = 1. \quad \checkmark$$

Assume it is true for $n = k$

$$\text{i.e. } 1(1)! + 2(2)! + 3(3)! + \dots + k(k)! = (k+1)! - 1$$

Try to prove true for $n = k + 1$

$$\begin{aligned} 1(1)! + 2(2)! + 3(3)! + \dots + k(k)! + (k+1)(k+1)! &= (k+1)! - 1 + (k+1)(k+1)! \\ &= (k+1)! \{1 + k + 1\} - 1 \\ &= (k+1)! (k+2) - 1 \\ &= (k+2)! - 1 \\ &= S_{k+1} \quad \checkmark \checkmark \end{aligned}$$

Hence if it is true for $n = k$, then it is true for $n = k + 1$

Now it is true for $n = 1$; hence it must be true for $n = 1 + 1$, i.e. $n = 2$.

If it is true for $n = 2$, then it is true for $n = 3$ and so on for all positive integers n $\checkmark$